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Swimming

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Humans are better built for walking and running than for swimming. If we compare the top speeds attained by the fastest running animal (the cheetah at 120 km/h) and swimming animal (the sailfish at 110 km/h), our fastest humans to date have reached 38% of that speed in running (Usain Bolt at 45 km/h with the 100 m track world record) and only 8% in swimming (Cesar Cielo at 8.6 km/h with the 50 m freestyle world record). Despite this fact, some anatomical and physiological adaptations suggest that human evolution has been more (semi-) aquatic than originally thought (14). In more modern times, swimming has developed into a popular recreational activity, becoming a competitive sport in Great Britain in the 19th century and part of the first modern Olympic Games in Athens in 1896.

This chapter outlines performance factors of the sport, the role that high-intensity interval training (HIIT) plays in developing athletes, snapshots of training schedules of swimmers who have won medals at the Olympic Games and/or World Championships, and practical methods to structure, plan, and monitor swim training.

Sport Description

The most obvious difference between swimming and land-based sports is that swimming is performed in the water medium, where reducing drag becomes

critical for success. (The drag on the human body moving in water is about 780 times greater than the drag involved with moving through air.) Toward minimizing drag, swimming is performed with as much streamline as possible, in the horizontal position, where propulsion comes through a continuous movement of the arms, in coordinated combination with the kick.

The competition frequency for an elite swimmer is not as regular as in most popular sports (e.g., weekly for soccer, several grand slams each year for tennis, two or three competitions per week for basketball and ice hockey). For swimmers, the highest level of competition is the Olympic Games, held every 4 yr in a long-course pool (50 m). Figure 17.1 shows an overview of the individual pool-swimming events of the 2020 Tokyo Olympic program, with respective world records (updated August 2018). Pool-event durations range from ~21 s in the men's 50 m freestyle to ~15 min in the 1500 m freestyle. Events not listed in the figure are the 10 km open-water race, which takes ~2 h, and the team relays (4 × 100 m medley, 4 × 100 m freestyle, 4 × 200 m freestyle, and mixed women/men 4 × 100 m medley relay). Each event comes with rules and regulations that are published at www.fina.org.

The competition environment at major championships is generally well controlled. The pool is always exactly 50 m for long-course competitions and exactly 25 m for short-course competitions; the

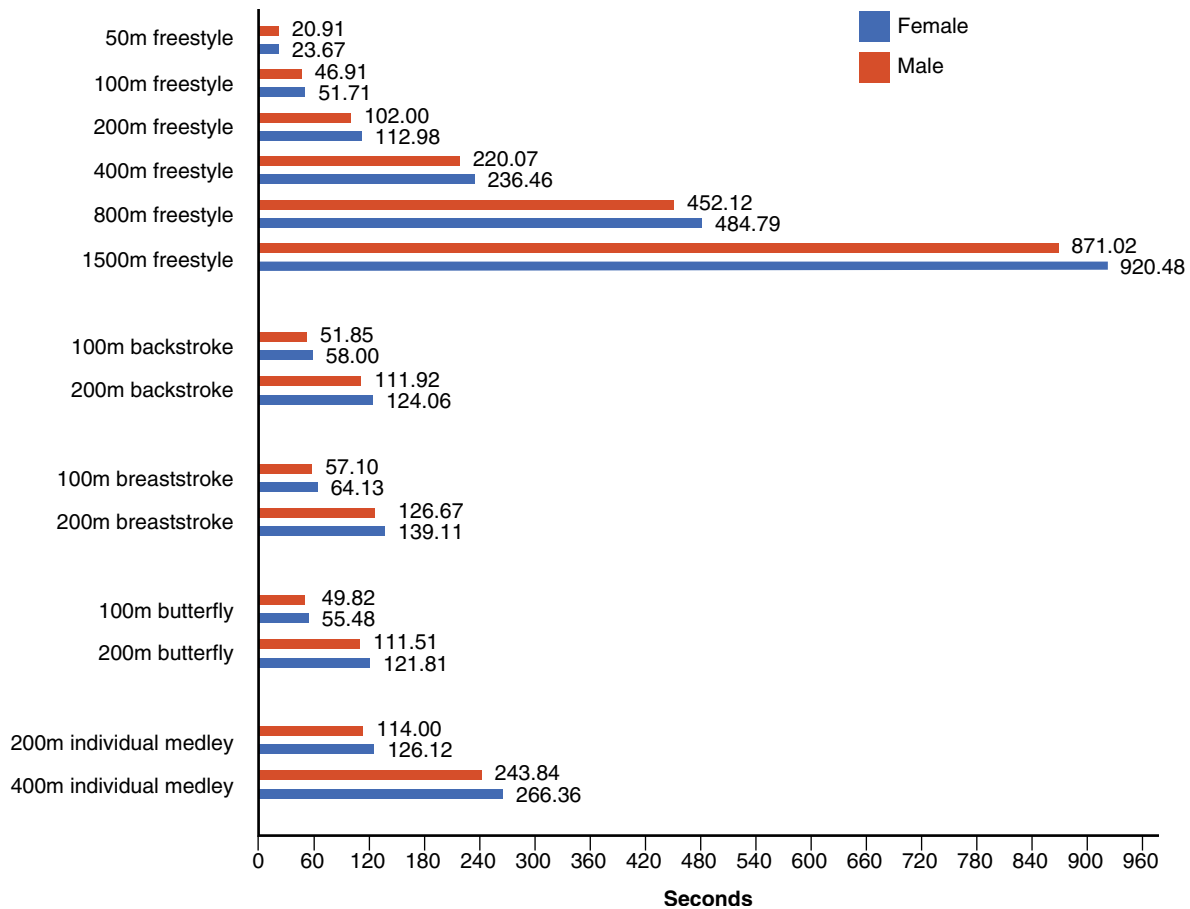


Figure 17.1 Individual pool events of the 2020 Tokyo Olympic program, with respective world records (updated August 2018) in male and female athletes. Record times are shown in seconds.

water temperature is standardized at $\sim 27^{\circ}\text{C}$; and pools used for Olympic Games or World Championship events have a minimum depth of 2.0 m. Olympic Games and World Championships are always held indoors; the last Olympic Games and World Championships that were held outdoors were in Athens in 2004 and Rome in 2009, respectively. Open-water events are clearly less controlled, with differences in weather, water current, water temperature, and swimming course.

At the World Championships and Olympic Games, the 16 best performances from heat swims in each event go through to semifinals, from which the eight best performances go through to the final, where the medals are won. Exceptions include the 400 m, 800 m, 1500 m, and relay events, where there are no semifinals, and the open-water events, where there is a single mass start in a straight final.

There are strict regulations regarding swimsuits. In 2008 and 2009, technologically advanced polyurethane swimsuits appeared in the sport, which made a

swimmer's body more buoyant and streamlined. Twenty world records were set at the 2009 World Championships in Rome as a result. In 2010, the polyurethane swimsuits were banned, after an American proposal to return to textile-only suits was passed with 180 nations voting for the ban and only 7 against it.

Physical Contributors to Performance in Swimming

Performance is multifaceted in any sport. When it comes to winning medals at major championships, superior physical fitness and technical skills must go hand in hand with confidence, belief, and the ability to perform under pressure. Training the mental side of performance and appropriate daily decision-making (e.g., appropriate rest, sleep, nutrition) is equally important as training the physical side.

Figure 17.2 places swimming among other sports in terms of different proportions of physical requirements of speed, strength, and endurance. Note the

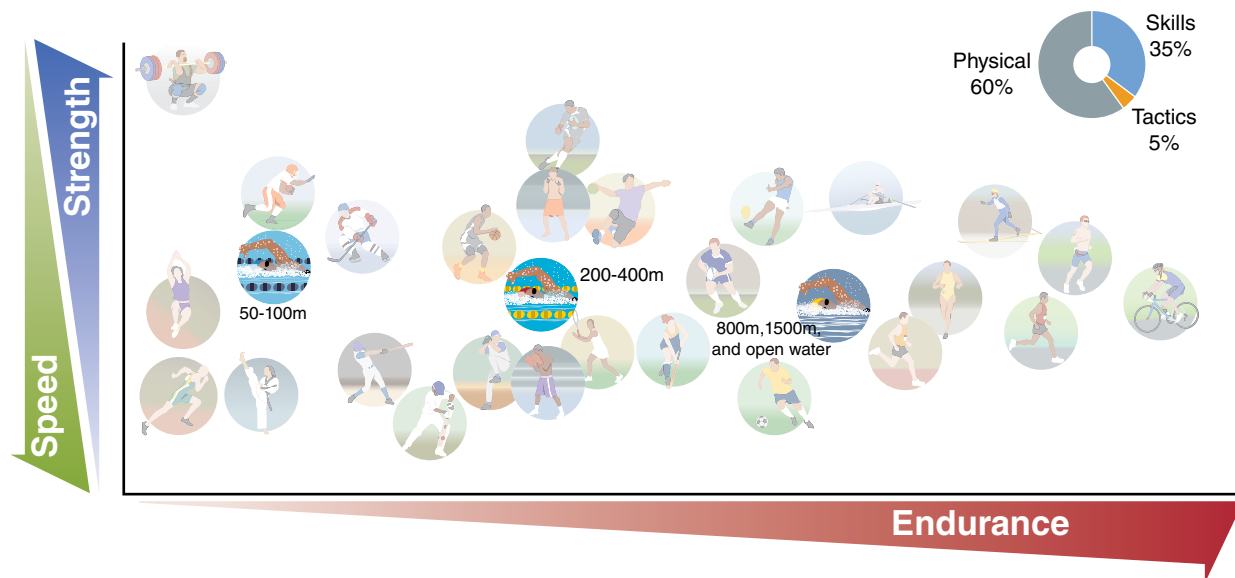


Figure 17.2 The position of the swimmer on the three axes will vary depending on the specificity of the event, with longer events naturally requiring more endurance, though other variables such as technique and propulsive efficiency may be even more important in the sport of swimming. Note the distinct position placement on the figure of swimmers specializing in the sprint (50-100 m), middle distance (200-400 m), and distance (800-1500 m and open-water) events.

Adapted from G.A. Nader, "Concurrent Strength and Endurance Training: From Molecules to Man," *Medicine & Science in Sports & Exercise* 38, no. 11 (2006): 1965-1970.

distinct position placement on the figure of swimmers specializing in the sprint (50-100 m) middle distance (200-400 m), and distance (800-1500 m and open-water) events. However, other factors such as technique, propulsive efficiency, and the ability to repeat performance under pressure are also important, demonstrated by the fact that some elite athletes win across a range of events (strokes and distances). The most prominent example is one of history's most successful Olympians, Michael Phelps (USA), who has won 23 Olympic gold medals and has set world records in backstroke, freestyle, individual medley, and butterfly events across different distances.

The physiological demands of swimming are different for each event, due to the large range in duration between them (figure 17.1). Figure 17.3 shows computer-simulated percentage energy contributions of the three energy systems involved in Olympic pool freestyle events in elite male athletes (39). The three relevant energy systems include the phosphagen system (anaerobic alactic), the glycolytic system (anaerobic lactic), and the aerobic oxidative system (see chapter 3). As shown in the figure and explained in chapter 4, the longer the duration of the event, the larger the contribution of the oxidative system; the shorter the duration of the event, the higher the

contribution of the glycolytic system. For simulations of metabolic responses during high-intensity interval training, we refer to the work of Hellard and colleagues (22).

Coaches will aim to maximize and balance the capacities of the energy systems, building those metabolic capacities with effective swim technique and neuromuscular patterns. We refer to chapters 3, 4, and 5, and specifically for the sport of swimming, to the work of coach/coach scientists Forbes Carlile (7), James Counsilman (10), Ernest Maglischo (28), Iñigo Mujika (31, 32), and Brent Rushall (40), and of scientists David Costill (9), Philippe Hellard (18, 19, 21, 22), Jan Olbrecht (33), and David Pyne (35, 37).

In addition to muscular metabolic capacities and neuromuscular patterns, the cardiac output, discussed in chapter 3, may be a discriminating factor. Many muscles are used during whole-body swimming exercise, resulting in a competition for blood flow (42). An increase in blood volume is therefore a desired training adaptation, as well as an increase in muscle blood flow capacity (26).

Anthropometric variables and body composition affect performance. While a swimmer's anthropometric characteristics are somewhat fixed after maturation (i.e., stature), body composition will

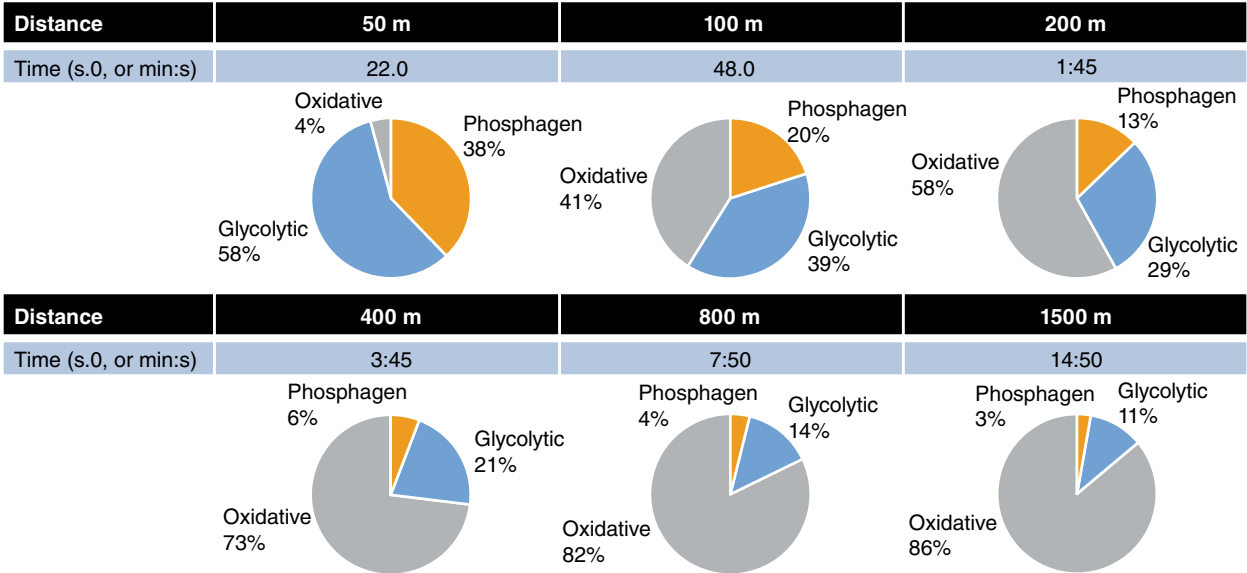


Figure 17.3 Share of energy system contribution during competitive freestyle swimming events in top-level swimmers using computer simulation.

Data from F.A. Rodriguez and A. Mader, “Energy Systems in Swimming,” in *World Book of Swimming: From Science to Performance*, edited by L. Seifert, D. Chollet, and I. Mujika (New York: Nova Science, 2011): 225-240.

change according to manipulation of training load and nutritional intake. For example, as training load increases, female middle-distance swimmers generally lose total body mass and body fat (skin-folds) with minimal changes in lean mass, while male middle-distance swimmers generally gain body mass and lean mass, but show minimal changes in skinfolds (3). Anderson and colleagues also observed a decrease in body fat (skinfolds) in the taper phase in female swimmers from one season to the next, which correlated strongly with improved competition performance. For male swimmers, increased proportional lean mass occurred during several phases of the training season, which also correlated strongly to improved end-of-season competition performance time (2).

Technical changes in swimming often result in larger performance gains compared to increases in metabolic power (12, 45). The coach’s experienced eye is an important vehicle used to drive such change. Technical demands vary by event, and breaststroke is a highly specialized event in this regard. Certain skills are also relatively more important in specific events. For example, the start is far more important in the 50 m freestyle than in the 1500 m freestyle. Other strategies that reduce drag include the use of different swimsuits (8), swim caps (to remove drag from head hair), and shaving (which in top male

swimmers is generally done only at major competitions and their selection events).

Another important performance factor in swimming is pacing. Athletes must pace themselves to get the best outcome, which is one of the more difficult aspects of competitive swimming, and is of importance in all swim events. For example, going out too fast in the first 50 m lap of a 100 m event can cost excessive amounts of energy, leading to a reduction in speed in the latter part of the race. Alternatively, not going out fast enough can leave a swimmer out of the race. An example in the 100 m freestyle may be that of Kyle Chalmers at the 2016 Rio Games. Kyle sat in seventh place with a 23.14 s split time at the 50 m turn, and was able to maintain velocity more effectively, returning in a 24.44 s split to win gold.

Pacing profiles do vary by event distance and by stroke (30). Swimmers typically adopt a *positive* pacing profile in all 100 m events, whereby lap time increases from the first to the second lap, which is largely due to the rapid acceleration obtained from the dive start. The most successful 100 m swimmers are able to maintain their speed more effectively in the final lap (38). In events 200 m and above, the most commonly used pacing strategies are called *fast-start even*, where laps are evenly paced after a fast first lap, and *parabolic*, which is similar but with a fast end-sprint in the final lap(s) (25, 29). Slight differences

have been observed in 200 m breaststroke and butterfly events with a positive pacing profile due to the unique characteristics of these strokes (38).

Targets of Physical Performance in Swimming

Coaches generally tend to categorize swimmers into four groups as follows:

1. Sprint swimmers, who train for 50 and 100 m events,
2. Middle-distance swimmers, who train for 200 and 400 m events,
3. Distance swimmers, who train for 800 and 1500 m events, and
4. Open-water swimmers, for those racing in 5, 10, and 25 km events in lakes, rivers, seas, and oceans.

To visualize the common differences in training between groups, a case-study snapshot is shown of a mid-season training week in an athlete who has won a medal in the respective event at the Olympic Games and/or World Championships. The reader will need to place these week snapshots in perspective, bearing in mind that

1. snapshots show the overall session mileage and training focus, not the actual HIIT content or the way (technique) the athlete executed the HIIT,
2. the athletes completing such training will have been training consistently for many years,
3. training weeks, mileage, focus, and HIIT content will change substantially according to the time of season and from one season to the next,
4. different training strategies may have impacted performance positively or negatively, and
5. other medal-winning athletes may have been through very different training strategies.

The training focus of sessions in these single-week snapshots are shown simply as low aerobic, high aerobic, anaerobic, race-specific, speed, and recovery, defined as follows:

- Low aerobic: swimming at speeds ~70 to 40 beats below maximum heart rate, or at ~75% to 85% of the maximum 400 m swimming speed;
- High aerobic: longer volume sets of swimming at ~85% to 110% of the maximum 400 m swimming speed;

- Anaerobic: training sets that usually consist of ~25 to 50 m repetitions swum very fast with appropriate passive rest to ensure a high contribution of anaerobic glycolysis;
- Race-specific: a speed specific to the individual's main event;
- Speed: very fast swimming up to ~20 m;
- Recovery: a session aimed at recovery, usually following a previous session involving a high training load. This session will consist of very easy swimming, generally no longer than 45 min.

Focus categories will be explained further with examples of training sets in the HIIT Weapons and Their Manipulations section.

Sprint

The sprint group trains to perform in 50 m and 100 m events.

50 m Events

For 50 m events, the competition performance duration is between ~21 s for elite male freestylers and ~29 s for elite female breaststrokers.

There are not many programs that specialize in 50 m events because (currently) the only Olympic event at this distance is the 50 m freestyle. At the 2016 Olympic Games in Rio, the performance difference between the 50 m freestyle Olympic champion and silver medalist was 0.01 s (or 0.05%) and 0.02 s (0.08%), respectively, for male and female athletes. Conversely, the 2017 World Championships showed an average 1.0% difference between gold and silver in 50 m events (in freestyle, backstroke, breaststroke, and butterfly, which are currently part of the World Championships programs). This higher percent difference is mainly explained by the exceptional performances of Great Britain's Adam Peaty and Sweden's Sarah Sjöström, who were 0.52 s (2.0%) and 0.78 s (3.2%) ahead of the silver medalists in the 50 m breaststroke and 50 m butterfly, respectively.

To be successful in 50 m events, a swimmer needs explosive power, with superior swim-specific power and skill. Although a small share of the energy delivery during a 50 m event is provided through the oxidative system (figure 17.3), most swimmers specializing in this event follow a combination of speed, anaerobic and aerobic training, and a larger volume than would be expected in comparison with similar-duration land-based events (e.g., track running).

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
6–8 AM	Aerobic and speed	Off	Technique	Off	Max speed and easy speed	Aerobic hypoxic	Off
	4000 m		4000 m		Type 5 3000 m	4000 m	
9–10 AM	Weights	Land training	Weights	Land training	Weights	Land training	
4–6 PM	Kick and transfer	Aerobic hypoxic	Speed range	Aerobic hypoxic	Kick and resistance	Off	Off
	Type 2 4500 m	4500 m	Type 5 3000 m	4500 m	Type 2 3400 m		
Total ~35,000 m							

Figure 17.4 Mid-season training week snapshot including swim-training focus and session distance of the program of an elite 50 m freestyle swimmer.

Figure 17.4 shows an example of a mid-season training week in a swimmer who has won an Olympic medal in the 50 m freestyle. The overall swimming volume of this particular week was 35 km. The week included six land-based training sessions, of which three were weight-training sessions; there is traditionally a relatively large emphasis on strength training in 50 m specialist programs.

100 m Events

In 100 m events, the event duration is between ~47 s for elite male freestyle and ~65 s for elite female breaststroke.

The 100 m long-course events consist of swimming two 50 m laps. While ~30% to 45% of the energy during a 100 m race is of an anaerobic glycolytic origin (19, 39), Hellard and colleagues have observed that only 5% to 8% of training is actualized and targeted as such for top French sprinters (20). Observations in Australian swimming programs are similar. The reason for this relatively low proportion of anaerobic training is twofold. First, there is a likely decrease in propulsive efficiency during high-intensity work in the water medium, where the coach will often stop the athlete when technique fails. Second, technique is considered critical to success, and such work has a relatively large aerobic component. The relatively large training volumes performed by swimmers will be discussed further in the HIIT Weapons and Their Manipulations section.

Competition performance times alongside personal observations in training indicate that some athletes may be more 50 and 100 m specialists, while others may be more 100 and 200 m specialists. The 50 and 100 m specialists usually have more

speed and a need for longer recovery times from high-intensity training (if performance in the particular subsequent session is the desired outcome), while the 100 and 200 m specialists usually handle more training volume and recover faster from high-intensity work. Swim training will always consist of some combination of speed, anaerobic and aerobic work, and it is up to the coach to decide how the volume and content of interval training will be applied and how adjustments are incorporated for an individual swimmer.

Figure 17.5 provides a snapshot of a mid-season training week in an elite 100 m freestyle swimmer. For all swimmers, it is important to develop their aerobic and anaerobic capacities with good technique. As such, it is important to develop and fine-tune both speed and race-specific skills. Figure 17.5 shows three weight-training sessions in this particular week. While there is much internal debate on the content of land-based work for swimmers, there is little clarity on how strong a swimmer needs to be for success. While land-based strength work can influence swim performance greatly in certain cases, the individualization and periodization of this training component is essential.

Middle Distance

The middle-distance group includes swimmers training to perform in 200 m and 400 m events.

200 m Events

In 200 m events, the durations are between ~1:45 for elite male freestylers and ~2:20 for elite female breaststrokers.

A 200 m event consists of four 50 m laps, with three turns. An elite 200 m swimmer will need to

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
6–8 AM	Low aerobic 5000 m	Low aerobic 2600 m	Off	Low aerobic 3300 m	Low aerobic 3600 m	200 and 100–m race pace Type 4 4500 m	Off
9–10 AM	Weights		Weights		Weights		
4–6 PM	High aerobic Type 2 4800 m	100-m race pace and speed Type 5 4800 m	Low aerobic 5000 m	High aerobic and race pace Type 4 5000 m	Low aerobic 3400 m	Off	Off
Total 42,000 m							

Figure 17.5 Mid-season training week snapshot including swim-training focus and session distance of the program of an elite 100 m freestyle swimmer.

maintain speed for a relatively long duration. As such, speed, anaerobic power, and aerobic power all need to be developed, maximized, and fine-tuned in training. Figure 17.6 shows the computer-simulated evolution of the metabolic contributions of the three energy systems in a real race situation for a 200 m freestyle Olympic swimmer (17). As with all forms of locomotion, the three energy systems are used simultaneously, though proportionally different, depending on the individual swimmer's system capacities, the duration and execution of the race, and the pacing strategy. Figure 17.7 shows a week snapshot of mid-season training of an elite 200 m swimmer. The schedule shows a range of focuses in training, aimed at developing aerobic and anaerobic capacities, and race-specific skills.

400 m Events

In 400 m events, the duration is between ~3:43 for elite male 400 m freestyle and ~4:30 for elite female 400 m individual medley.

The time to make a final at the 2016 Rio Olympic Games in the 400 m freestyle was 3:45.43 and 4:04.36 for male and female athletes, respectively. Very often, top 400 m swimmers are also elite in other distances, but either in the 200 and the 400 or in the 400, 800, and 1500 m events. This observation compares with the different aerobic versus anaerobic dominant 1500 m middle-distance running groups described in chapter 13. An exception in swimming is American Katie Ledecky, who is reigning Olympic Champion in the 200, 400, and 800 m events, and World Champion in the 1500 m (there was no female 1500 m event at the 2016 Olympics).

Figure 17.8 shows a week snapshot of mid-season training in a 400 m freestyle Olympic medalist. This

particular training week contained 10 swim sessions, of which 4 were high-aerobic training sessions, with an intended rating of perceived exertion of 8 or 9 out of 10, 4 low-aerobic sessions, and 2 recovery sessions. Land-based work consisted of 2 sessions enabling maintenance of already developed strength and a boxing session at the very end of the week.

Distance

Distance events include the 800 m and 1500 m events. Event durations are between ~7:30 in male 800 m freestyle and ~15:30 in female 1500 m freestyle.

Distance swimmers' training programs generally consist of a high training volume. Figure 17.9 shows the realized total swimming volume of an elite distance swimmer in the year leading to an Olympic Games. This figure shows volume only, which shouldn't be interpreted as training load, as training load will depend on how this mileage was realized (intensity $\times$ volume; chapter 8) with the relative physiological and psychological stress imposed (response to load; chapter 9). There are programs that keep the quality (swimming at higher intensities) relatively low and balanced, but there are also many successful programs that have a lot of quality in the majority of sessions throughout the week. Distance swimmers do not have the speed of sprinters and can't achieve the same absolute intensities as sprinters, which may make it possible for them to recover faster and to sustain more quality work for longer. The energy used during 800 and 1500 m races is mainly produced through aerobic oxidation (e.g., figure 17.6). Nevertheless, anaerobic capacities also need to be developed to be able to deal with changes in pace during competition, if the coach and athlete desire development of such a trait.

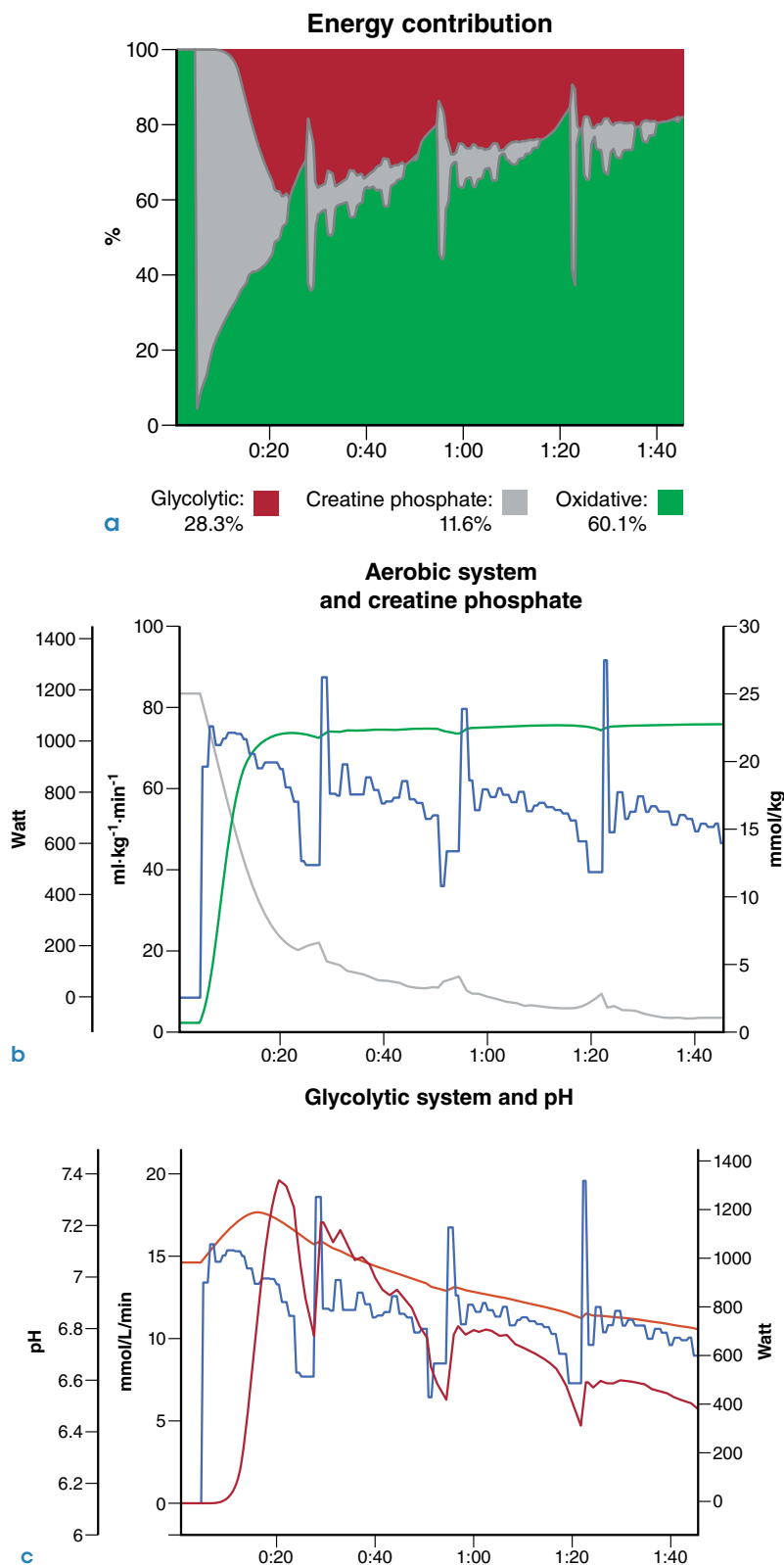


Figure 17.6 Computer simulations of relative energy contributions during the course of a 200 m race in an elite swimmer (modeled for a performance time of 1:46.99) (15). (a) % contribution of the three relevant energy systems for the duration of the race; (b) creatine phosphate (mmol/kg, gray line), $\dot{V}O_2$ (mL.kg⁻¹.min⁻¹, green line), and swim power (Watt, blue line); (c) rate of glycolysis (mmol/L/min, red line), level of pH (values below 7 indicate acidity that increases as the number decreases, orange line), and swim power (Watt, blue line).

Reprinted by permission from P. Hellard, S. Weber, F. Rodriguez, and A. Mader, "Influence of Oxidative and Glycolytic Capacities on the Muscular Energy Metabolism and Performance in 200-m Freestyle Swimming: A Case Study," presented at 4th FINA World Aquatics Convention, Windsor, Canada, 2016.

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
AM	Low aerobic to threshold 6000 m	Low aerobic 6000 m	Off	Low aerobic 5200 m	Low aerobic 6000 m	Race pace Type 4 5000 m	Off
AM	Weights		Weights		Weights	Land training	
PM	High aerobic Type 2 5000 m	Anaerobic Type 5 4000 m	Low aerobic and speed 5500 m	High aerobic and race pace Type 4 5500 m	Low aerobic and speed 5000 m	Off	Off
Total 53,200 m							

Figure 17.7 Mid-season-week snapshot including swim-training focus and session distance of the program of an elite 200 m freestyle swimmer.

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
AM	Low aerobic kick-pull 7000 m RPE 5	Low aerobic 7500 m RPE 4	High aerobic Type 2 7000 m RPE 8	Low aerobic kick-pull 7000 m RPE 5	Low aerobic and anaerobic set Type 5 8000 m RPE 6	High aerobic Type 2 6500 m RPE 8	Off
		Weights (maintenance)			Weights (maintenance)	Boxing session	
PM	High aerobic and speed Type 2 6000 m RPE 9	Recovery and speed 4500 m RPE 2	Off	High aerobic and speed Type 2 6000 m RPE 9	Recovery 4000 m RPE 2	Off	Off
Total 63,500 m							

Figure 17.8 Mid-season training week snapshot with swim-training focus, session distance, and intended rating of perceived exertion (RPE) of training in an elite 400 m freestyle swimmer.

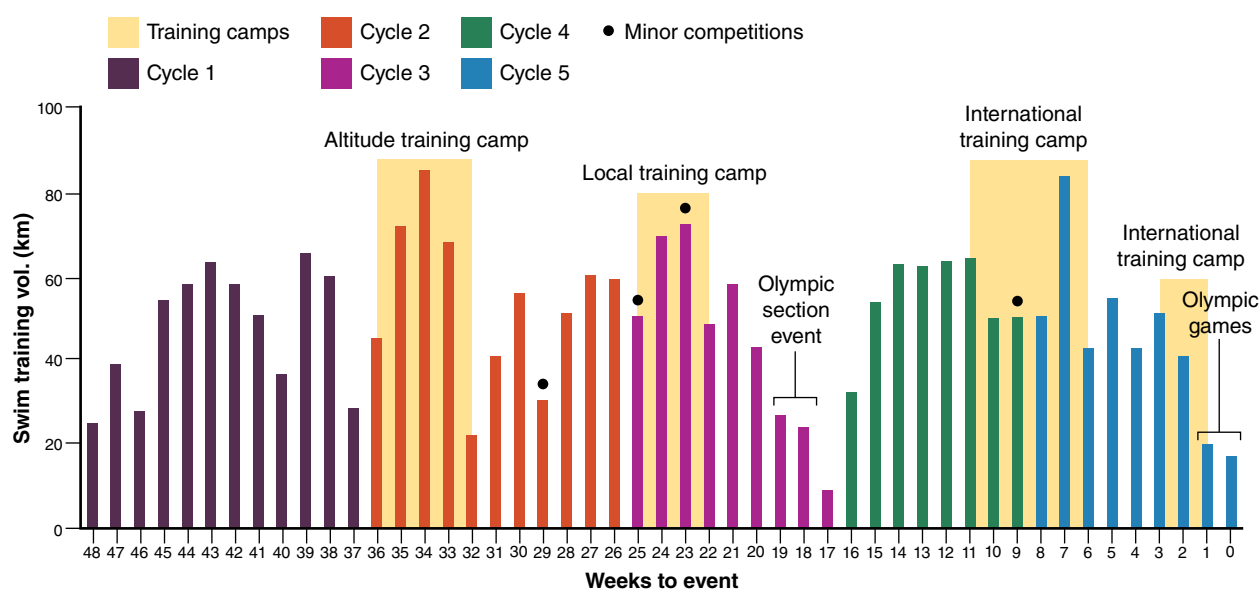


Figure 17.9 Realized swim training volumes (km) of an elite distance swimmer in an Olympic year. The X-axis shows the number of weeks until the start of the Olympic Games.

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
AM	Low aerobic 7200 m	Low aerobic 7000 m	High aerobic threshold 7000 m	Low aerobic 7000 m	Low aerobic and speed 7000 m	High aerobic 800-m reps 6000 m	Recovery 3000 m
	Weights (maintenance)		Weights (maintenance)		Cardio 30 min		
PM	Low aerobic and speed 7000 m	High aerobic 50-m reps Type 2 5500 m	Off	800, 400, 200 race pace Type 4 6000 m	Low aerobic 6500 m	Off	Off
	Total 69,200 m						

Figure 17.10 Mid-season training week snapshot with swim-training focus and session distance of the training program of an elite distance swimmer.

Figure 17.10 shows a week snapshot of training in an elite distance swimmer. This particular week included 11 swim sessions, with four high-aerobic sessions, six low-aerobic sessions, and one session on Sunday aimed at recovery and continuity in training. The Tuesday evening session was a session with 24×50 m repetitions swum at high intensity, but with relatively short rest. The volume of this high-intensity work (1200 m) and the relatively short rest cycles made the overall training set high aerobic (and not anaerobic). The two weights sessions were aimed at maintaining already developed strength.

Open-Water Swimmers

The open-water Olympic event is 10 km, a race that takes about 2 h; open-water races are conducted in lakes, rivers, or seas.

At the 2016 Rio Olympic Games, Dutch swimmer Ferry Weertman touched just ahead of Greece's Spiros Gianniotis to win Olympic gold. Both touched the finish mark in a time of 1:52:59.8, separated by a few hundreds of a second. France's Marc-Antoine Olivier finished ~2 s later for bronze. This example demonstrates that open-water swimmers need endurance, but also speed and anaerobic power, as a race is very often won in the final surge or at the final touch.

Figure 17.11 shows the changes in speed during a particular open-water race in an elite athlete, measured with an accelerometer. There was substantial change in pace in this particular race, of which some can be explained by differences in current. It does demonstrate though that open-water swimmers need to be able to deliver and deal with changes in pace effectively. Other important performance factors are

orienting skills, management of water and weather conditions, and ability to navigate buoys and fence off other competitors. Swimming too wide off the buoys could result in ~10 m more swim distance, for example. The swimming conditions can be rough, with stories of goggles slammed off, ankles groped, and heads punched. Referees try to keep an eye on this, but it's often difficult for them to see it all. There are feeding stations at certain points on the course, where support staff can reach out to provide liquid foods, so deciding when to feed is part of the individual's race strategy. The drinking bottles in the figure show where the respective athlete fueled during the race.

Figure 17.12 shows a snapshot of a training week of an elite female open-water swimmer. This training-week example included a large volume of high-aerobic work with speed variations and acceleration work, to mimic intrarace speed variations. The total swim volume of the week was ~79 km, realized in 10 sessions.

Key Weapons, Manipulations, and Surveillance Tools

As throughout the book, the weapon term refers to the HIIT formats that can be used to target certain physiological responses (figure 1.5), and the surveillance tools are what is used to monitor training load (chapter 8) and individual responses (chapter 9).

HIIT Weapons and Their Manipulations

Always keep in mind that swimming is a very technical sport. Any intensity needs to be swum with

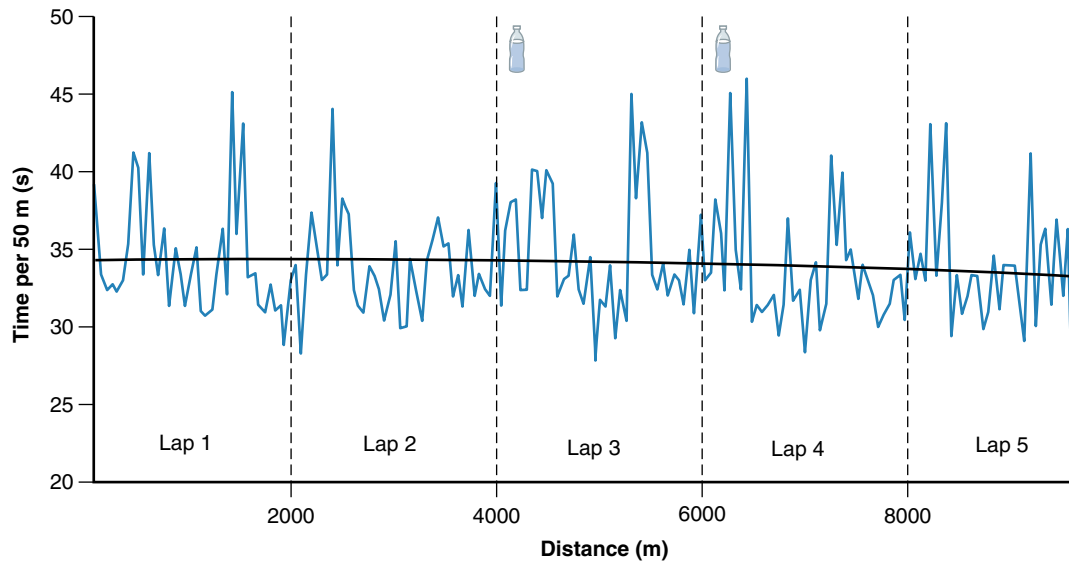


Figure 17.11 Changes in swim pace that occurred during a particular 10 km open-water race in a top athlete, measured with an accelerometer. The drinking bottles represent the timing where the athlete received food from a feeding station.

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
AM	High aerobic 100-m intervals	Low and high aerobic 200 and 400-m intervals	Anaerobic and high aerobic 50 and 300-m intervals	Low aerobic	Anaerobic and high aerobic 75-m intervals	High aerobic 100 and 300-m intervals	Off
	Type 1 7000 m	7500 m	Type 4 8800 m	7400 m	Type 4 6800 m	Type 4 8200 m	
	Weight circuit	Running		Weight circuit			
PM	Low and high aerobic 1000-m intervals	Low and high aerobic 250-m intervals	Off	High aerobic 800-m intervals	Low and high aerobic 400-m intervals	Off	Off
	8500 m	7500 m		8500 m	8400 m		
Total 78,600 m							

Figure 17.12 Mid-season-week snapshot with training focus, interval distance of the main training set, and overall swim-session distance of a training program in an elite female open-water swimmer.

great technique. Energy needs to be applied effectively toward forward movement, while reducing drag is equally crucial for achieving this effect. Throughout sport, coaches often use the word *quality* to describe the intensity of a workout, but in swimming (and probably other sports as well), it should be interpreted as intensity with great technique and skill. Neuromuscular, aerobic and anaerobic adaptations for swimmers all need to be built on optimal technical skill and execution.

Swim training volumes (figures 17.4, 17.5, 17.7, 17.8, 17.10, and 17.12) are typically high relative to the duration of the competition event (figure 17.1).

Elite athletes usually swim in the pool for 2 to 6 h each day, combined with land-based work, appropriate nutrition, and rest. One possible reason for these relatively high training volumes relates to the *time on task* needed to engrain and adapt the neuromuscular blueprint that optimizes forward propulsion in the water medium. Another possible reason is that swimming is a relatively slow continuous movement. For example, male 800 m running versus 200 m swimming have comparable world records of 100.91 s and 102.00 s, respectively. The ground/water-contact time (duration of force production), however, is ~0.1 s per stride in running versus ~0.6 s

per arm stroke in swimming. As a result, there is substantially more time in swimming to create propulsive force per stroke. Neuromuscular characteristics like tone, stiffness, and peak power are therefore relatively less important in swimming, while aerobic oxidative contribution to performance becomes relatively more important. Another possible factor for the higher training volumes of swimmers relates to the lower mechanical impact that swimming places on bones and joints compared to running and team sports, which makes it possible to train more.

While the large discrepancy between the physiological demands of training versus competition can be difficult to support for some physiologists, the majority of coaches observe success with these higher volumes of work, and to our knowledge, the vast majority of swimmers who have won Olympic medals have had a background of relatively large swim training volumes.

Observations in Australian and French swim programs are that some programs will avoid high-aerobic work and use a more polarized training schedule consisting of high-intensity work and predominantly low-intensity work (43), while other programs incorporate substantial volumes of high-aerobic work. Whether one strategy or another works best, or at what point in the periodized plan or career progression a specific strategy works best, largely depends on characteristics of the individual athlete, the event the athlete is training for, and the coach's specific beliefs and philosophical framework. As well, certain terms have specific meaning depending on the context; specific race-pace work for distance swimmers is high aerobic, while for sprinters it is more anaerobic.

Generally, most of the swimming volume (~60%–90%) is performed at intensities clearly below the anaerobic threshold; other volume is swum as HIIT. Paton and Hopkins (34) have published a good review on the effects of high-intensity training on performance. They classified HIIT as supramaximal, maximal, and submaximal intervals, as well as resistance exercise (including explosive, plyometrics, and weights). There are of course many ways to categorize training (see chapter 8). As already defined in this chapter, training is categorized into simple terms that include *low aerobic*, *high aerobic* (may include type 1, 2, 3, or 4 targets), *anaerobic* (type 3, 4, or 5 targets), *race-specific* (predominantly type 2 for distance swimmers, types 2 and 4 for middle-distance swimmers, and types 4 and 5 for sprint

swimmers), *speed* (type 5), and *recovery* to refer to the main areas of focus for each session. For the purposes of this chapter, the focus is on swim-specific training, omitting land-based training from discussion, although resistance and other land-based training methods form part of most successful swim training programs.

To further explain categories and swim-specific HIIT weapons, examples of training sets have been included, with indicative performance times and rest cycles of elite swimmers performing those sets in a long-course pool. Performance times will vary from swimmer to swimmer, and will depend on the current fitness and level of fatigue of the swimmer. Performance times are usually substantially faster in short-course pools, as experienced swimmers benefit from the push-off in the turns. Freestyle is the fastest stroke, followed by butterfly and backstroke; breaststroke performance times are substantially slower.

Low Aerobic

Low-aerobic training is performed at speeds ~70 to 40 beats below maximum heart rate, or at ~75% to 85% of the maximum 400 m swimming speed. This low-aerobic training intensity category is broad in swimming, with a relatively large range of heart rate intensities. Swimming longer repetitions below *threshold*, but still relatively fast, will predominantly use the aerobic system. With *threshold*, we refer to the anaerobic threshold, which is meant to indicate the maximal sustainable training speed without a net systemic H^+ accumulation. It is the maximum pace that a swimmer can maintain for longer distances/repetitions (see related descriptions in chapters 3 and 4). When athletes are training at threshold intensities, they induce a high stress on the oxidative system (16). The expected adaptations from performing low-aerobic work on a regular basis include metabolic and structural (enhancing $\dot{V}O_{2\max}$ and increasing the economy of swimming at $\dot{V}O_{2\max}$), in addition to the aforementioned neuromuscular remodeling. An example of this type of work for distance swimmers might be 6×800 m repetitions on a 10 min cycle. Top female distance swimmers would swim each repetition in around 9:30 min:s (~1:11/100 m) at relatively low heart rates, so they'd have about 30 s rest before starting the next 800 m repetition. Less fit or untrained swimmers would not manage this 10 min interval, and if they did, they would swim these times at much higher intensities, with higher anaerobic glycolytic engagement

and overall training stress. Middle-distance swimmers do this type of work regularly as well, but usually with shorter repetitions, e.g., 6×300 m repetitions ($\sim 1:10/100$ m) on a 4 min cycle, which would allow them ~ 30 s rest depending on the exercise intensity, or 8×200 m repetitions ($\sim 1:10/100$ m) on a 2:30 cycle, which would allow them ~ 10 s rest. Some coaches standardize the recovery interval, e.g., a fixed 30 s rest after each repetition, but most coaches fix the interval in practice (e.g., on a 4 min cycle, meaning that each repetition starts 4 min after the previous repetition began), as programs with groups of swimmers are easier to manage this way.

High Aerobic

High-aerobic training uses longer volume sets of swimming at $\sim 85\%$ to 110% of the maximum 400 m swimming speed. The high-aerobic training intensity bandwidth generally consists of high-volume, short-distance, and short-rest repetitions swum at near maximum, as well as longer distance repetitions swum as fast as possible.

High Volume, Short Distance, Short Rest Training of a high volume of shorter distance repetitions, with short rest, is swum near or at maximum *best-average* pace. (Best-average means that the average time of all repetitions needs to be as fast as possible.) An example of this short-interval work would be 30×100 m freestyle repetitions at *threshold pace* on a 1:20 cycle. This threshold-pace cue translates to fast swimming, though at a sustainable pace and with good technique. Top male and female distance freestyle swimmers would swim these 100 m repetitions at ~ 59 s and ~ 63 s with about 20 or 15 s rest, respectively, before starting the next 100 m repetition. This set could also be swum at a higher intensity, e.g., 30×100 m freestyle repetitions best-average on a 1:40 cycle, where top male and female distance freestyle swimmers would swim the 100 m repetitions at ~ 56 s and ~ 60 s, respectively, so they'd have about 40 s rest before starting the next 100 m repetition (type 2). Variation of training intensity is important, and HIIT can be manipulated by changing the distance of intervals, the overall distance of the set, the rest interval, and/or the intensity of swimming (see also chapter 4).

Longer-Distance Repetitions Swum as Fast as Possible An example of this long-interval work involves 5×400 m repetitions at best-average speed on a 5 min cycle. Top male distance swimmers would

swim $\sim 4:10$ min:s for each repetition; top female distance swimmers $\sim 4:22$ min:s. That means that male distance swimmers would get ~ 50 s rest, and female swimmers ~ 38 s rest, between repetitions.

The anticipated adaptations from doing this high-aerobic work on a regular basis are also metabolic and structural (enhancing $\dot{V}O_2\text{max}$ and increasing the economy of swimming at $\dot{V}O_2\text{max}$), and neuromuscular. The difference between these intensities and a swimming coach's low-aerobic intensity prescription is that there will be a higher stress on both the oxidative and glycolytic systems (types 3 and 4). This type of exercise provides a high training load, and usually warrants a relatively longer recovery time in the subsequent period.

Anaerobic

Anaerobic training consists of ~ 25 to 50 m repetitions swum very fast, with a high contribution of anaerobic glycolysis. An example of this work, which has aspects of repeated sprint training and sprint interval training weapons, and hits type 4 targets, is 4×25 m dive max and 4×35 m dive max with appropriate passive rest in between repetitions; top male swimmers would swim the 25 m repetitions in ~ 10 s and the 35 m repetitions in ~ 15 s.

An example of type 4 repeated sprint training targeting high-level stimulus of both anaerobic glycolysis and aerobic oxidation is 10×50 m butterfly at 100 m race-pace stroke rate, swum as fast as possible on a 2 min cycle, with passive rest between repetitions. Figure 17.13 shows the relevant physiological data of a female Australian international-level swimmer who has performed this training set. The total volume of the set, 500 m, is relatively high, which made the overall training stimulus more aerobic. Active recovery between the repetitions would also make the training set more aerobic (see chapters 4 and 5).

In the example (figure 17.13), the athlete's average 50 m performance time was 30.4 s, resulting in ~ 90 s rest between repetitions. This swimmer's 100 m race-pace stroke rate was 51 strokes/min, with the goal of swimming as fast as possible at this prescribed stroke rate for each 50 m repetition (see related training methods in chapter 16, rowing). As soon as the swimmer touched the wall at the finish of each repetition, a Cosmed K_4b_2 portable metabolic analyzer was worn for ~ 45 s to estimate $\dot{V}O_2$ during exercise using back extrapolation (11). The data showed that a $\dot{V}O_2$ peak was achieved by the 6th repetition with

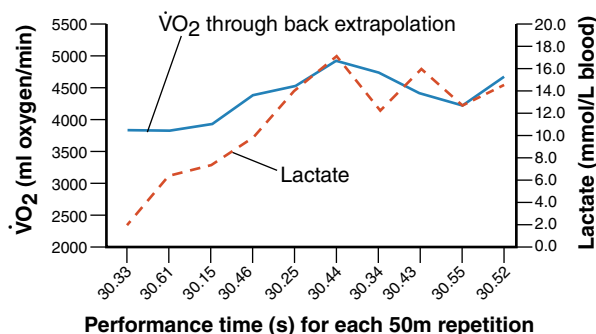


Figure 17.13 Performance data, with $\dot{V}O_2$ obtained through back extrapolation and blood lactate concentrations, of a female Australian international-level 200 m butterfly swimmer performing 10 $\times$ 50 m butterfly repetitions at a stroke rate of 51 cycles/min on a 2 min cycle (meaning each repetition starts 2 min after the previous repetition began). This session highlights the HIIT type 4 target.

this high-intensity work (type 4). The data have not been represented as a percentage of the swimmer's $\dot{V}O_{2\max}$, as assessment of $\dot{V}O_{2\max}$ remains challenging in swimming.

Typical adaptations from this aerobic and anaerobic metabolic signaling during high-intensity swimming include an increase in the aerobic oxidative and anaerobic glycolytic metabolic contributions (27), an increase in buffering capacity, which translates to an improved ability to swim with the most efficient combination of stroke rate and stroke length during competition, and an increase in neuromuscular coordination at fast swim speeds. The training sets can be manipulated through changing the total number of repetitions, the rest interval, the distance of each repetition, and the stroke rate, for example (see also chapters 4 and 5). Many coaches will have the athlete stop this specific work when technique is failing, though some coaches will have the athlete push through, to challenge the swimmer to maintain good technique and fast swimming under fatigue.

Race-Specific Training

Physiologically, race-specific work for sprint swimmers could also be categorized *anaerobic* (type 4 and 5 targets), while for middle-distance swimmers it could be *anaerobic* (type 4) or *high aerobic* (type 2), and for distance and open-water swimmers *high aerobic* (predominantly type 2 targets). Training the specificity of competing in a certain event is clearly

important (figures 1.5, 17.1, and 17.2), not only for developing the necessary physiological capacities but also for the technical skills, the mental skills, the pacing, and maintenance of great technique under fatigue.

To train with specificity, some coaches and athletes work from a performance model, which typically includes target performance times for the start (usually the first 15 m of the race), the turns (usually measured from 5 m into the wall through to 10 m off the wall), the laps (50 m laps in a long-course pool), with stroke counts and rates (strokes per minute). Table 17.1 shows an example performance model for a top 200 m individual medley swimmer. Aspects of this model can then be tested and trained for.

An example of a race-specific training set for 100 m sprinters is 2 $\times$ (60 m dive at 100 front-speed execution, followed by a 50 m push effort at execution of the second lap of the race), with $\sim$ 40 s between the 60 m dive and the 50 m push efforts, and appropriate recovery and reactivation between the two sets. An example of a race-specific training set for a 200 m middle-distance swimmer is 2 $\times$ (75 m dive at 200 m get-out pace swim easy through to the 100 m mark, $\sim$ 10 s rest, 3 $\times$ 50 m push efforts at 200 pace with $\sim$ 20 s rest in between), with appropriate recovery and reactivation between the two sets. An example for 1500 m distance swimmers might be 20 $\times$ 100 m on a 1:20 cycle at 1500 m race-pace ($\sim$ 58 s/100 m for male swimmers). The anticipated stimulus and adaptations are similar to those described in the anaerobic work section, except are specific to the individual's race-pace motor pattern. Coaches and athletes also often use minor domestic and international competitions for training purposes, to practice very specifically for competition and to fine-tune racing skills, including the mental component.

Speed Work

Speed work consists of $\sim$ 10 to 20 m repetitions at very fast speeds with relatively long rest, triggering the phosphagen system. An example of speed work in a swimmer, which resembles the repeated sprint-training weapon hitting type 5 targets, might be 4 $\times$ 20 m maximal sprint dives on a 2 min cycle. Top male freestyle athletes would swim $\sim$ 8 s on each repetition, meaning they'd have $\sim$ 110 s rest between repetitions. With speed work, the rate and pattern of muscle fiber stimulation by the central nervous system is increased, and the phosphagen system is triggered. This work is also aimed at developing and

Table 17.1 Individualized 200 m Individual Medley Performance Model for a Total Time of 1:56.99 (min:ss.00)

A 200 m individual medley consists of four 50 m laps, swum as butterfly, backstroke, breaststroke, and freestyle by lap.

Stroke	Segment	Segment split time (s)	Lap split time (s)	Strokes per lap (n)	SR (strokes/min)	In (s)	Out (s)	Turn (in + out, s)
Butterfly	0-15 m	5.95						
	0-25 m	11.22			52			
	25-50 m	13.58	24.80	18	51	2.71	5.16	7.87
Backstroke	50-75 m	14.35			42			
	75-100 m	15.10	29.45	34	42	3.01	5.94	8.95
Breaststroke	100-125 m	16.82			39			
	125-150 m	17.65	34.47	17	39	3.36	5.62	8.98
Freestyle	150-175 m	14.02			46			
	175-200 m	14.25	28.27	38	46	2.61		

SR: stroke rate; In: time between head passing the 5 m mark into the wall to the touch on the wall; Out: time from the touch of the wall through when the head passes the 10 m mark off the wall; Turn: sum of in and out as an overall 15 m turn time.

fine-tuning technical speed skills in individual athletes. The distance of each repetition typically ranges between 10 and 30 m, though the longer the distance and the shorter the rest, the more the anaerobic lactic and aerobic systems become involved. Manipulations can be achieved through changing the number of repetitions, rest cycles, and distance, or, for example, by including a turn off the wall.

Recovery Swimming

The most important recovery modalities for swimmers are rest, sleep, and nutrition. Discussion around recovery is beyond the scope of this chapter and is covered in chapter 7 in detail. Factors that will affect the need for recovery include the training intensity and volume, psychological stress, health and fatigue status, and the environment (e.g., temperature, humidity, altitude). A recovery swim session will generally consist of very easy swimming, which is usually no longer than 45 min or 2 km. The examples in figures 17.8 and 17.10 show recovery sessions with higher mileages (~3-4000 m), mainly because coaches usually want distance swimmers to achieve a certain mileage at the end of the week.

Load Surveillance and Monitoring Tools

The measurement and monitoring of training load, discussed in chapter 8, is an important issue for athletes, coaches, and sports scientists. A number

of informative reviews on the quantification of training have been produced (15, 24, 41). Here, we will mention a few methodologies used regularly in the sport of swimming to prescribe and monitor the external and internal training load in swimmers. As expanded on in chapter 8, the external training load involves the work completed by the individual, while the internal training load refers to the relative physiological and psychological stress imposed by the external training load.

Prescription and Monitoring of Exercise Intensity in Training Sessions

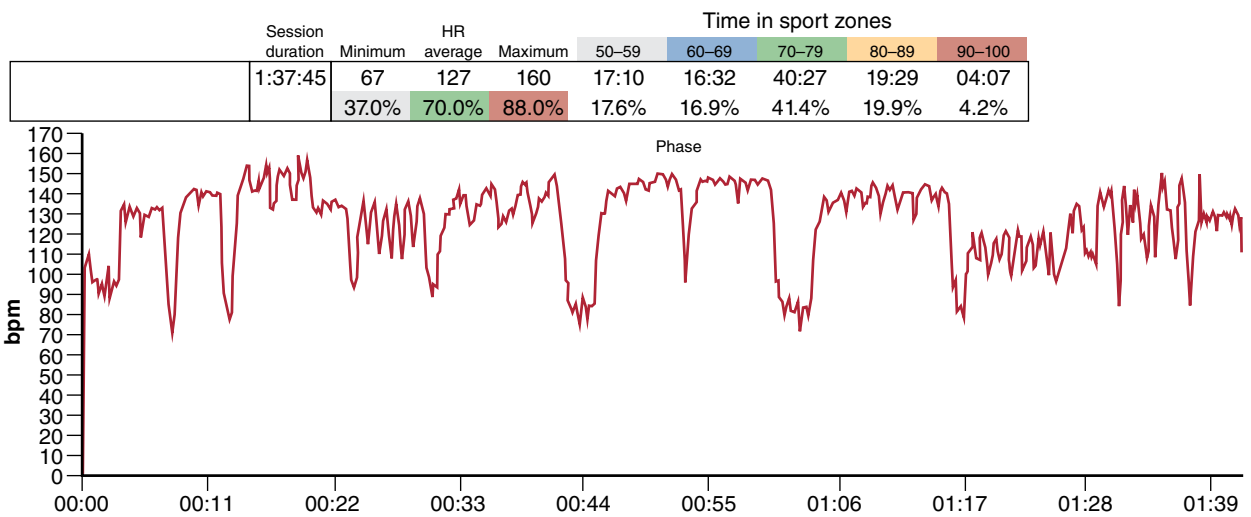
Swimming coaches often prescribe the desired training intensity based on the *heart rate* associated with submaximal efforts. In reality, however, this beats below maximum heart rate prescription in fact is a prescription of *perceived effort*. The measuring and monitoring of heart rate in swimming remains challenging, and there are also issues of validity. As discussed in chapters 2, 8, and 9, heart rate is not a valid prescription or assessment of anaerobic and speed work, or for high-intensity work swum at near maximum heart rate. There is also the issue of the validity of using heart rate to prescribe exercise intensity across individual swimmers. While heart rate shows an almost linear relationship with $\dot{V}O_2$ at submaximal intensities, this relationship is individual (1). The use of heart rate zones (chapter 8) for both prescription and monitoring of submaximal loads also needs assessment of the individual's

maximum heart rate during swimming, so that beats below maximum can be both prescribed and monitored. The assessment of an individual's maximum heart rate will require a pool-based incremental and/or repeat effort test to volitional exhaustion. To our knowledge, there are no gold-standard protocols used in the literature for swimming research.

There are various devices and systems available on the market to measure heart rate in swimmers; some are used to measure heart rate at the end of each repetition, while other devices have been developed to monitor heart rate throughout exercise. The challenge with measuring heart rate at the end of a repetition includes the time lag of measurement, the time needed for the measurement (especially when there is little rest between intervals), and the effect of pacing on heart rate, when a last lap was swum faster, for example. The challenge of monitoring heart rate throughout each repetition in swimming resides in the capturing of data and ensuring its quality. Most devices capture data with a chest strap, and while this can be put underneath the swimsuit for

female swimmers, male swimmers need to swim with vests, which they find restrictive. The straps also move from time to time, especially in push-offs or dive starts. There are companies working toward capturing heart rate from the temporal vein (through integrated sensors in goggles or swim caps), but we have yet to see this properly functional. Figure 17.14 shows an example training session of a top distance swimmer, where heart rate was measured throughout using a Polar Team 2 System. The heart rate data follow the intensities according to the work in the low-aerobic training session nicely, but too often the data is not clean enough for the system to be used as a load-monitoring system in low-aerobic and high-aerobic swimming.

A method that is regularly used to prescribe training intensity is setting a *goal pace* (desired performance time) for repetitions. This pace can be in relation to any training intensity: easy aerobic swimming, race-pace swimming, maximal efforts, and speed. The advantage of using this method is that performance time can be easily measured and



- 4 x 300-m freestyle (breathe every 3 and every 5 strokes by 50-m lap) on a 5-min cycle
- 4 x 100-m medley, change strokes at the 25-m mark on 1:45 cycle
- 1 x 200-m freestyle drill on 4-min cycle
- 6 x 50-m freestyle, work on distance per stroke, on 1-min cycle
- 4 x 200-m freestyle paddles pull on 2:45 cycle (HR 130-150)
- 2 x 400-m kick on 8-min cycle (HR < 150)
- 10 x 100-m freestyle or backstroke by 50-m lap on 1:30 cycle
- 8 x 50-m freestyle drill with fins on 1-min cycle
- 8 x 50-m medley order as (20 m fast/30 m recovery) on 1:15 cycle
- 2 x 600-m choice swim

Figure 17.14 Example of a training diary of an athlete specializing in 400 m and 800 m freestyle events. HR: heart rate; RP: race pace; Flow: how you felt moving through the water. Monitoring permits data to be shown as changes on a week-by-week basis.

monitored in the pool; the disadvantage is that the realized intensity at a certain training pace will be individualized and depends on the levels of fatigue and efficiency.

Variables that are used less frequently to prescribe intensities are the *stroke rate* and *RPE* (5). Figure 17.13 provides an example of how intensity can be prescribed using stroke rate, where the swimmer's instruction was 10 × 50 m butterfly repetitions at 51 strokes per min on a 2 min cycle. There are devices that can be used to help with standardizing a stroke rate (e.g., Finis Tempo Trainer). This tool works with some athletes, but other athletes find the beeping sound too much of a distraction to achieve good rhythm and technique. In any case, most experienced swimmers are aware of what stroke rate they swim at. The RPE method, described in chapter 2, has been validated as a robust and reliable measure for submaximal intensities (13), though many coaches prefer cues such as easy swimming, very fast swimming, race-pace swimming, maximum effort swimming, or beats below maximum heart

rate, for example. Figures 17.8 and 17.14 have portions of the prescription where RPE and other cues are used to prescribe the session intensity.

Monitoring Training Load

Figure 17.9 shows the swim training volumes of an elite distance swimmer in an Olympic year. While the figure nicely describes the weekly training distances realized, it does little to describe the intensities of this work, and therefore isn't a valid measure of training load. Automated systems using inertial sensors have more recently become available to measure swimming speed, stroke rate, and mileage. While these systems have yet to properly monitor the external swim training load, good gains have been made in this area, and useful systems are predicted to be available in coming years. Currently, coaches and scientists often dissect training mileage into mileage swum at certain training intensities, which has its challenges, but provides a more accurate measurement of external training load. Monitoring tools used to assess internal training load, described in

World Championships, days to go...		93	92	91	90	89	88	87
		Mon	Tue	Wed	Thu	Fri	Sat	Sun
		22 May	23 May	24 May	25 May	26 May	27 May	28 May
Sleep	Total hours of sleep	8	8	7	8	7	8	10
	Hours of sleep during day	1	1		1			
	Quality of sleep (see scale)	7	7	7	8	7	7	8
AM training session	Low aerobic ($\leq$ HR 155, m)	5900	6700	5900	7640	6400	4400	
	High aerobic ($>$ HR155<RP, m)	1200	300				3000	
	800-m race pace (m)						300	
	400-m race pace (m)							
	200-m race pace (m)					150		
	100-m race pace (m)					100		
	Speed (0-25 m, m)			100	160	50		
	Total mileage (m)	7100	7000	6000	7800	6700	7700	0
	Training intensity (see scale)	6	6	4	5	7	8	
	Flow (see scale)	5	4	5	5	5	6	
Gym	AM gym training intensity			7				
PM training session	Low aerobic ($\leq$ HR 155, m)	3280	6525		5500	6600		
	High aerobic ($>$ HR155<RP, m)	3600			900	400		
	800-m race pace (m)							
	400-m race pace (m)							
	200-m race pace (m)		475		600			
	100-m race pace (m)							
	Speed (0-25 m, m)	120						
	Total mileage (m)	7000	7000	0	7000	7000	0	0
	Training intensity	8	8		8	6		
	Flow	7	6		6	5		
Gym	PM gym training intensity	7						

Flow (how you feel moving through the water)

Sleep quality

Training intensity

10 Extremely good
9 Very, very good
8 Very good
7 Good
6 Fairly good
5 Average
4 Fairly poor
3 Poor
2 Very poor
1 Very, very poor
0 None

10 Extremely hard
9 Very, very hard
8 Very hard
7 Hard
6 Fairly hard
5 Average
4 Fairly easy
3 Easy
2 Very easy
1 Very, very easy
0 No training

Figure 17.15 Example of a training diary of an athlete specializing in 400 m and 800 m freestyle events. HR: heart rate; RP: race pace; Flow: how you felt moving through the water. Monitoring permits data to be shown as changes on a weekly basis.

Figure 17.16 Example of three macrocycles for a sprint swimmer building toward the 2018 Commonwealth Games and Pan Pacific Games, showing mesocycles with training focus and target week swim mileage. AEC: aerobic capacity; ANC: anaerobic capacity; REG: regeneration; Comp Prep: competition preparation. The main competitions are in yellow; the minor competitions are in orange. The global structure of two of these mesocycles is expanded in detail in figure 17.17.

Figure 17.16 Example of three macrocycles for a sprint swimmer building toward the 2018 Commonwealth Games and Pan Pacific Games, showing mesocycles with training focus and target week swim mileage. AEC: aerobic capacity; ANC: anaerobic capacity; REG: regeneration; Comp Prep: competition preparation. The main competitions are in yellow; the minor competitions are in orange. The global structure of two of these mesocycles is expanded in detail in figure 17.17.

Regeneration	Days out	22 Jan	23 Jan	24 Jan	25 Jan	26 Jan	27 Jan	28 Jan	36,000
	AM	Low aerobic 4000	Low aerobic 4500	Low aerobic 4500	Low aerobic 5000	Recovery 3000	Low aerobic 5000	Off	
			Land circuit	Land circuit		Land circuit	Land circuit		
	PM	Off	Speed 2500	Off	Contrast Fast vs. easy 5000	Speed 2500	Off	Off	
Comp Preparation	Days out	29 Jan	30 Jan	31 Jan	1 Feb	2 Feb	3 Feb	4 Feb	35,500
	AM	High aerobic 5000	Low aerobic 5000	Low aerobic 5000	Contrast Fast vs. easy 5000	Recovery 3000	Anaerobic 3000	Off	
		Land circuit	Land circuit		Land circuit	Land circuit			
	PM	Speed 2500	Anaerobic 3000	Off	Race pace 4000	Off	Off	Off	
Comp Preparation	Days out	5 Feb	6 Feb	7 Feb	8 Feb	9 Feb	10 Feb	11 Feb	35,500
	AM	High aerobic 5000	Low aerobic 5000	Low aerobic 5000	Contrast Fast vs Easy 5000	Recovery 3000	Anaerobic 3000	Off	
		Land circuit	Land circuit		Land circuit	Land circuit			
	PM	Speed 2500	Anaerobic 3000	Off	Race pace 4000	Off	Off	Off	
Taper	Days out	12 Feb	13 Feb	14 Feb	15 Feb	16 Feb	17 Feb	18 Feb	27,500
	AM	High aerobic 4000	Low aerobic 4000	Low aerobic 4000	Speed 2000	Recovery 2000	Anaerobic 3000	Off	
		Land circuit	Land circuit		Land circuit	Land circuit			
	PM	Speed 2000	Anaerobic 2500	Off	Race pace 4000	Off	Off	Off	
Taper	Days out	19 Feb	20 Feb	21 Feb	22 Feb	23 Feb	24 Feb	25 Feb	22,500
	AM	High aerobic 3000	Low aerobic 3000	Low aerobic 3000	Speed 2000	Recovery 2000	Anaerobic 2500	Off	
		Land circuit	Land circuit		Land circuit	Land circuit			
	PM	Speed 2000	Anaerobic 2000	Off	Race pace 3000	Off	Off	Off	
Competition		26 Feb	27 Feb	28 Feb	1 Mar	2 Mar	3 Mar	4 Mar	18,600
	10:30 AM	Light swim 2000	Race prep 2500	Race prep 1600	100 free heat 2000	Recovery 2000	50 free heat 1500	Low aerobic 2000	
	7:30 PM	Speed 2000	Off	Off	100 free final 2000	Off	50 free final 1000	Off	

Figure 17.17 Example overview of a 5 wk training block of a 100 and 50 m specialist swimmer, working toward a competition, showing overall swim training volumes and focus. This training block aligns with two of the mesocycles shown in figure 17.16. Training focus is categorized into simple terms that include *low aerobic*, *high aerobic* (could include type 1, 2, 3, or 4 targets), *anaerobic* (type 3, 4, or 5 targets), *race-specific* (predominantly type 2 for distance swimmers, types 2 and 4 for middle-distance swimmers, and types 4 and 5 for sprint swimmers), *speed* (type 5), and *recovery*; *contrast* is a session including very fast swimming and easy swimming (in line with type 4 targets).

chapter 8, include RPE and heart rate, of which session RPE may be current best practice. The use of session RPE also allows for the simultaneous monitoring of land-based training loads, which in effect add substantially to the overall load and are not included in the monitoring of swim training volumes.

Various swim squads around the world use training diaries, logbooks, or digital applications to monitor certain variables of load. Figure 17.15 shows a snapshot of an athlete's training diary, with variables of external training load (realized mileage at certain intensities per training session) and internal training load (perceived training intensity), as well as other monitoring variables such as hours and quality of sleep, and training flow (how the swimmer felt moving through the water). The challenge of the use of training diaries lies in the validity of the entered data, though they can provide good information for training and performance evaluation.

A promising method to characterize training load during high-intensity training and competition is the simulation of metabolic energy responses using mathematical models (17, 19, 22, 39). This method shows promise to help coaches and scientists better understand, visualize, and monitor the metabolic response and cost of high-intensity training and competition in individual athletes.

Strategies for Structuring the Training Program

Coaches will generally aim to have their athletes reach peak performance at major championships and major championships-selection event for their country (as dictated by the National Swimming Organization), where the athlete will need to achieve

qualifying standards set by their respective organizing committee. The uncontrollable factors that the athlete and coach must navigate include the race date, venue, event order, and timing of major championship events relative to the selection trial events. Athletes and coaches will work with the competition calendar framework, using available time as effectively as possible to achieve their goals. Figure 17.16 is an example of a competition calendar that extends from August 2017 to August 2018, including competitions, training camps, and mesocycles with physiological focus and weekly target mileages. Medical checks, skinfold assessment, testing sets, training blocks, and so on can all be included within the yearly plan. Once this overview is established, the coach can then schedule the weekly microcycle training structures, deciding when and how to schedule HIIT work throughout the respective week.

Figure 17.17 shows a 5 wk training block of an athlete working toward a selection event for a major competition. This training block aligns with two of the mesocycles outlined in figure 17.16, and shows target swim volumes with individual session focuses. Examples of HIIT training sets according to the training focus have been described previously in this chapter. While the planning process is developed with the aim of reaching peak performances at certain times of the year, it unfortunately does not guarantee that maximal performance potential will be expressed in competition. As discussed previously, performance is a manifestation of many variables, which makes the pursuit of solving the performance puzzle, through coaching and the scientific support of individual athletes, a fascinating and evolving journey. It is as much an art in achieving focus on what is important in preparation for individual swimmers as it is an art for the athlete to perform when it matters most.

THINKING OUTSIDE THE BOX

Inspiration and opportunity have brought us to work with some of the world's best swimmers and their coaches, and to gain insights into training structures, HIIT content, and coach philosophies of top swim programs in Australia, France, the Netherlands, United States, New Zealand, Sweden, Denmark, and Spain.

In swimming, applied physiologists generally focus on monitoring performance in HIIT sets, including technical (e.g., stroke rate and stroke count) and physiological (e.g., RPE, lactate, heart rate) measures, to drive and assess change, and to bring individualization to training. As described in the chapter, warranted adaptations and individualization in training is mainly done through periodizing and individualizing the distance and the number of HIIT repetitions, the rest between the repetitions, the time between HIIT sessions, and the execution of HIIT.

In physiology, we also try to look within, e.g., on a muscular, cardiovascular, or hormonal level, and that's often difficult to do—more so than, for example, in biomechanics, where angles and technique can be assessed more directly through video analysis, or in anthropometry, where height, skinfolds, and bone lengths can be measured straightforwardly.

Methods to look within are, for example, the measurement of $\dot{V}O_2$ during or immediately after swimming (11, 23, 44), blood lactate profiling (33), and step tests (36). Another topical method to look within is the collaboration between radiologists and sports scientists in the Muscle Talent Scan, a method used to noninvasively estimate muscle fiber-type composition in athletes using an MRI scanning process. Here, the measurement of certain metabolites, such as carnosine concentration, can help to discriminate between fast-twitch, intermediate, and slow-twitch athletes, which may be useful for showing event-specific performance potential (4).

Wim, Philippe, and Tom have different roles in Belgium, France, and Australia, respectively, but all three are part of helping athletes and coaches achieve their goals, of helping people make more-informed decisions, and of contributing to the existing scientific knowledge. The three of us are part of a team in respective nations, and occasionally get together as a team internationally, to attempt to answer questions of coaches collectively, to address specific parts of the performance puzzle together, and in this occasion to write a book chapter. We believe our jobs are among the best in the world and are very grateful for the privilege. We hope that this chapter will help aspiring swim sport scientists and coaches think about what they do, and that it will help to structure their thinking into delivering successful training strategies. In line with the theme emphasized throughout the book, we believe our collaboration demonstrates “thinking outside the box” in applied sport science (6).

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